

Weighted Population Estimates

Analysis of poverty.dta using census-adjusted expansion weights

7,185,103

Estimated household population

28,740,504

Estimated individual population

This report presents weighted population totals derived from 9,600 household-level records. The estimates expand the observed records to the represented population using the supplied 2021 census-adjusted household weight. They are population estimates, not counts of records in the sample.

Variables Used

Variable	Analytical role
psu_number	Primary sampling unit identifier; paired with hh_number to identify each household uniquely.
hh_number	Household identifier within the primary sampling unit.
hhs_wt	Household expansion weight adjusted to the 2021 census; used to estimate the number of represented households.
hhsiz	Number of people in the household; used with the household weight to calculate the household's contribution to the individual total.
ind_wt	Person expansion contribution. It was validated against hhs_wt * hhsiz within numerical tolerance.

The combination of psu_number and hh_number was confirmed to be unique across the 9,600 observations. All variables required for the calculations were nonmissing and positive, so no household records were excluded from either total because of missing or nonpositive analytical values.

Weight Calculation

Each household contributes its expansion weight, hhs_wt, to the household population estimate. Conceptually, a household with a weight of 500 represents 500 households in the expanded population. Summing this contribution once for every unique household produces the household total.

```
Estimated households = sum(hhs_wt) = 7,185,102.657485962
```

For individuals, the household expansion weight is multiplied by household size. Thus, a household's person-level contribution is `hhs_wt * hhsize`. The supplied `ind_wt` variable was checked and found equal to that product within tolerance, supporting its direct summation as the expanded individual population.

```
Estimated individuals = sum(ind_wt) = sum(hhs_wt * hhsize)
= 28,740,503.526668549
```

Methodology

The analysis treated `poverty.dta` as a household-level file. Before estimation, record structure was checked by testing uniqueness of the composite household key `psu_number + hh_number`. Required identifiers, weights, and household-size values were then checked for completeness and positivity. This validation establishes that each record enters the totals once and that the expansion calculations are based on usable values throughout the dataset.

Weighted totals were obtained by direct summation across the 9,600 validated household records. No additional aggregation, imputation, or adjustment was introduced. The household estimate is the sum of `hhs_wt`; the individual estimate is the sum of `ind_wt`, with the identity between `ind_wt` and `hhs_wt * hhsize` serving as a consistency check. Reported headline figures are rounded to the nearest whole household or person, while the full-precision sums are shown above for reproducibility.

Scope of inference: These calculations provide expanded totals only. PSU- and strata-based variance estimation was not performed, so this report does not present standard errors, confidence intervals, significance tests, or claims about design-based precision.

Interpretation

After application of the 2021 census-adjusted expansion weight, the 9,600 observed household records represent an estimated **7,185,103 households** and **28,740,504 individuals**, rounded to whole population units. The decimal values in the unrounded results arise because survey expansion weights are typically not integers; rounding is applied only for readable presentation of the final totals.

Dividing the individual estimate by the household estimate yields an implied weighted average household size of approximately **4.000 persons per household**. This relationship is internally consistent with the validated construction of the person contribution from household weight and household size. The estimates should be read as the total populations represented by the weighted data, not as 9,600 sampled households or as a statement about the uncertainty surrounding the totals.

Source: calculations from `poverty.dta`. Household weights are described in the data as adjusted to the 2021 census. Totals are rounded to the nearest whole unit for display.